

Automated Phishing Detection for Frontier AI Inference

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Design → evaluation → diagnosis
Modification → measured comparison
Complete research revision



The questions promised in August and September

RQ1: What incremental value do structural URL features and character-level representations provide under registrable-domain-disjoint and external evaluation?

RQ2: Can GMM-based monitoring detect an external source/domain shift and guide escalation without exceeding the low-FPR operating constraint?

RQ3: What detection, escalation, throughput, and latency tradeoffs determine whether the fixed cascade is viable inline?

Exact active wording: August 20 and September 3 decks; carried forward September 17.

Evidence coverage and population boundaries

What was measured

- Internal: 34,593 rows; 14,326 positives and 20,267 negatives.
- External: 8,701 retained rows; gold 69, certified 2,497.
- 125 operational cells; 250 owned service/client exits verified.

What the labels mean

- Publisher reference labels, not independent adjudication.
- Tranco: 1,163 label-free controls, never labeled FPR.
- 72 retained + 53 new cells; no accepted adverse repeat replaced.

URL-representation and two-session amendments remain disclosed; no claim of lifetime-unseen test data.

RQ1: internal recall does not ensure external specificity

Frozen detector	Internal recall	Internal FPR	Gold recall	Certified FPR
Length-only	34.61%	0.5872%	23.19%	9.4513%
Logistic-L1	98.68%	0.7747%	100%	90.5487%
Fixed cascade	98.68%	0.7747%	100%	90.5487%
Transformer	99.32%	1.0115%	100%	99.2791%

**Every primary external FPR gate fails.
Perfect gold recall is not evidence of a safe operating point.**

Denominators: internal P=14,326 / N=20,267; external gold P=69 / certified N=2,497.

RQ1: structural gain; no incremental fixed-cascade gain

Contrast	Recall gain (pp)	Clustered 95% interval (pp)
Internal L1 – length	64.07	[59.99, 67.63]
Internal cascade – L1	0.00	[0.00, 0.00]
Gold L1 – length	76.81	[66.67, 86.96]
Gold cascade – L1	0.00	[0.00, 0.00]

**The fixed band selects 0/34,593 internal and 0/8,701 external rows.
H1: not supported — 5 pass, 5 fail; all 10 measured.**

2,000 domain-clustered replicates; internal positives: 9,757 domains; gold: 69 domains.

RQ2: detecting departure did not make routing useful

Monitoring

- External alerts: $115/132 = 87.12\%$; passes $\geq 80\%$.
- Original audit: $28/252 = 11.11\%$; fails $\leq 5\%$.
- MMD and PSI: 132/132 external alerts; descriptive only.

Routing consequences

- Transformer routing: $8,253/8,701 = 94.85\%$.
- Gold recall gain: 0; interval $[0,0]$.
- Certified FP: $2,261 \rightarrow 2,479$; policy FPR 99.2791% .

H2: not supported — 1 pass, 3 fail. Overlapping windows do not establish independent harmful-drift events.

RQ3: low transformer use did not guarantee low latency

Primary operational gates

- Reference physical calls: 0/10,000; passes $\leq 30\%$.
- Concurrency-64 pooled p95: 364.30 ms; fails ≤ 200 ms.
- Errors: 0/50,000; passes $< 0.1\%$.

External safeguards

- Cascade certified FPR: 90.55%; transformer: 99.28%.
- Both alert on 1,163/1,163 Tranco controls.
- Gold noninferiority passes, but does not establish specificity.

H3: not supported — 3 pass, 5 fail. Primary cells were retained unchanged, not rerun for a better result.

The complete service matrix includes adverse load results

Group	Pooled p95 ms	Errors / requests	Physical calls
Fixed 1%, c1	1.49	0 / 50,000	0%
Fixed 1%, c64	364.30	0 / 50,000	0%
Transformer, c64	476.34	74 / 50,000	99.88%
Transformer, c128	970.86	1,612 / 50,000	96.964%
Live shift, c1	11.58	0 / 43,505	94.851%

All 25 groups and five repeats retained; full matrix in appendices.

Closed-loop throughput is not production capacity. Under-100% transformer attempts can reflect failures.

Secondary evidence qualifies—not repairs—the primary result

Completed coverage

- 153 population/model records; all seeds and comparators.
- 43 mixed-class metric sets; 430 calibration bins; 129 prevalence projections.
- Full low-FPR curves, McNemar/Holm, MMD/PSI and label-free probes.

Important qualifications

- RF certified FPR: 91.35%; no secondary model is promoted.
- Source-only fit is unidentifiable; permutation audit limits remain explicit.
- Seed 42 runtime differs; probes do not establish correctness or robustness.

No best-seed selection, post-test threshold replacement or fabricated negative-control success.

One engineering process, with distinct evidence stages

Design and evaluate

- Define URL, risk, routing and service contracts.
- Measure all 22 primary checks and 25 operational groups.
- Identify where component benefits meet or miss joint requirements.

Diagnose and compare

- Trace scheme spelling through structural features.
- Implement a transport-neutral representation; freeze its threshold.
- Measure an additional benchmark; isolate HTTP client topology.

Original H1–H3 remain unchanged. The bounded extension adds measured mechanism-level evidence, not rewritten history.

Diagnosis selects two bounded interventions

Representation pathway

- Scheme affects is_https, length, ratios and entropy.
- Dropping is_https alone does not remove scheme information.
- Use one modeling prefix for every feature; omit is_https.

Service pathway

- No-model diagnostics expose client/transport overhead.
- Compare shared versus worker-owned persistent clients.
- Keep service, scorer, inputs and request limits unchanged.

Diagnostics motivate tests; they do not establish a single cause. Transformer, GMM and routing are not retuned.

Detection comparison: freeze first, evaluate every admitted row

Additional benchmark

- Publisher 2020 benchmark: 11,430 source rows.
- 8,622 admitted: $P=4,651$; $N=3,971$; 6,273 domains.
- 2,808 exclusions; overlap and quarantine records retained.

Fixed comparison

- Unchanged 25-feature L1 comparator and threshold.
- 24-feature candidate: one fit; development-only calibration.
- All admitted rows scored once; paired domain bootstrap.

Retrospective, publisher-labeled comparison; shared upstream feeds. Domain exclusion is not temporal independence.

D: full operating-point tradeoff at the frozen thresholds

Metric	Unchanged L1	Scheme-neutral
Recall	99.31%	95.36%
False-positive rate	99.22%	92.90%
Precision	53.97%	54.59%
ROC AUC	0.6821	0.6950
Average precision	0.6651	0.6976
Brier score	0.4306	0.4077

251 fewer false positives; 184 fewer detected positives.

D: not supported — recall gain and the 1% FPR requirement are not met.

Recall difference -3.96 pp; 97.5% domain-cluster interval $[-5.05, -3.00]$ pp. No threshold retuning.

Achieved property: exact scheme invariance

Prescribed metamorphic check

- Swap HTTP ↔ HTTPS; keep the remainder unchanged.
- Candidate features and scores match on 8,622 / 8,622 pairs.
- Candidate decisions: zero flips.

What the comparison establishes

- Unchanged L1: 481 decision flips; 2,909 exact score matches.
- The modified feature path removes the intended dependency.
- Stable representation alone does not ensure low-FPR transfer.

Companions test representation, not additional labeled websites. Comparator flips are not automatically classification errors.

Measured service improvement; strict contract distinguished

Primary structural workload, c64

- Median paired p95 ratio: 0.20325 (79.68% reduction).
- 97.5% interval: [0.18790, 0.21351].
- Worker p95: 72.00 ms; zero errors / 100,000.

Prediction versus response order

- 99,999 / 99,999 comparable predictions agree.
- One shared timeout leaves one pair noncomparable.
- Strict agreement: 799 / 100,000; ordering differs.

Four of five S requirements pass. S is not supported as a conjunction; original H3 is unchanged.

All ten primary pairs: unchanged structural scorer, c64

Pair	Shared p95 ms	Worker p95 ms	Ratio
1	343.16	72.61	0.21158
2	371.79	67.27	0.18093
3	359.24	70.01	0.19488
4	358.86	72.86	0.20302
5	397.67	71.59	0.18003
6	342.30	74.72	0.21829
7	362.08	74.31	0.20524
8	352.65	71.76	0.20347
9	341.78	73.63	0.21543
10	353.73	71.29	0.20155

Every pair retained. Success-only p95; one shared timeout in pair 2 is retained separately. No best-run selection.

Controls locate the measured benefit in the client path

Workload	c	Client	p95 ms	Errors	Req/s
Control	1	shared	1.10	0	1039.9
Control	1	worker	1.11	0	1036.5
Control	64	shared	345.95	0	469.2
Control	64	worker	67.21	0	1647.6
Structural	1	shared	1.42	0	820.4
Structural	1	worker	1.43	0	819.8
Structural	64	shared	358.37	1	454.1
Structural	64	worker	72.00	0	1526.7

Ten arms and 100,000 attempts per row. Similar c1 latency; both c64 workloads improve. No transformer speedup claim.

Technical contribution: measured changes, explicit contracts

What the artifact demonstrates

- Structural representation: +64.07 pp internal recall.
- Scheme-neutral inputs: exact invariance on 8,622 pairs.
- Worker-owned clients: 79.68% paired p95 reduction.

What the evidence distinguishes

- Representation stability versus external low-FPR detection.
- Shift sensitivity versus useful escalation after the alarm.
- Prediction equivalence versus admission-order equivalence.

Reusable artifact and evaluation contracts; no invention claim for normalization, regression or connection reuse.

Completed research and dissertation evidence

Original commitments delivered

- All 125 cells, 25 operational groups and 22 primary checks.
- Direct RQ1–RQ3 answers; explicit H1–H3 decisions.
- Secondary analyses, raw-count tables and audit limits.

Iteration and final artifacts

- D: all 8,622 rows and companions; no retuning.
- S: all 80 arms; independent saved-evidence recomputation.
- GWU manuscript, editable deck, aggregates and provenance.

All adverse results and interruptions retained. Research completion is not institutional acceptance.

Appendix: every H1 gate

Gate	Operand	Required	Result
internal.length only.fpr	0.5872%	<= 1%	PASS
internal.logistic l1.fpr	0.7747%	<= 1%	PASS
internal.cascade.fpr	0.7747%	<= 1%	PASS
certified.length only.fpr	9.4513%	<= 1%	FAIL
certified.logistic l1.fpr	90.5487%	<= 1%	FAIL
certified.cascade.fpr	90.5487%	<= 1%	FAIL
internal.logistic l1 minus length only	59.9897 pp	> 0 pp	PASS
internal.cascade minus logistic l1	0.0000 pp	> 0 pp	FAIL
gold.logistic l1 minus length only	66.6667 pp	> 0 pp	PASS
gold.cascade minus logistic l1	0.0000 pp	> 0 pp	FAIL

Contrast operands are lower confidence bounds; all decisions use full precision. No missing gates.

Appendix: every H2 gate

Gate	Operand	Required	Result
external window alerts	87.1212%	$\geq 80\%$	PASS
audit window alerts	11.1111%	$\leq 5\%$	FAIL
certified.policy.fpr	99.2791%	$\leq 1\%$	FAIL
gold.policy minus cascade	0.0000 pp	> 0 pp	FAIL

Contrast operands are lower confidence bounds; all decisions use full precision. No missing gates.

Appendix: every H3 gate

Gate	Operand	Required	Result
certified.cascade.fpr	90.5487%	$\leq 1\%$	FAIL
certified.transformer.fpr	99.2791%	$\leq 1\%$	FAIL
tranco.cascade.alert rate	100.0000%	$\leq 1\%$	FAIL
tranco.transformer.alert rate	100.0000%	$\leq 1\%$	FAIL
gold.cascade minus transformer	0.0000 pp	≥ -2 pp	PASS
reference transformer invocations	0.0000%	$\leq 30\%$	PASS
http pooled p95 ms	364.3004 ms	≤ 200 ms	FAIL
http request errors	0.0000%	$< 0.1\%$	PASS

Contrast operands are lower confidence bounds; all decisions use full precision. No missing gates.

Appendix: full operational matrix (1/3)

Group (5 repeats)	p50 ms	p95 ms	p99 ms	Errors / requests
Fixed 1% c1	1.27	1.49	1.69	0/50000
Fixed 1% c8	5.75	10.03	13.46	0/50000
Fixed 1% c16	18.48	45.58	65.89	0/50000
Fixed 1% c32	43.10	130.18	196.80	0/50000
Fixed 1% c64	103.39	364.30	581.59	0/50000
Fixed 1% c128	237.49	897.34	1452.18	120/50000
Fixed 0.1% c1	1.26	1.48	1.65	0/50000
Fixed 0.1% c8	5.77	10.04	13.62	0/50000
Fixed 0.1% c16	18.70	45.35	65.70	0/50000

Warmup excluded; terminal latencies include failures. All individual throughput/counter records supplied.

Appendix: full operational matrix (2/3)

Group (5 repeats)	p50 ms	p95 ms	p99 ms	Errors / requests
Fixed 0.1% c32	45.83	138.86	212.35	0/50000
Fixed 0.1% c64	102.18	341.84	528.29	0/50000
Fixed 0.1% c128	222.65	795.94	1217.36	36/50000
Fixed 5% c1	1.27	1.49	1.65	0/50000
Fixed 5% c8	5.93	10.31	13.94	0/50000
Fixed 5% c16	18.14	44.86	68.40	0/50000
Fixed 5% c32	45.37	137.90	207.37	0/50000
Fixed 5% c64	102.13	338.41	518.37	0/50000
Fixed 5% c128	227.82	831.31	1291.33	59/50000

Warmup excluded; terminal latencies include failures. All individual throughput/counter records supplied.

Appendix: full operational matrix (3/3)

Group (5 repeats)	p50 ms	p95 ms	p99 ms	Errors / requests
Transformer 1% c1	7.14	8.08	8.58	0/50000
Transformer 1% c8	49.03	52.73	54.58	0/50000
Transformer 1% c16	98.30	104.86	107.31	0/50000
Transformer 1% c32	200.79	243.88	266.04	0/50000
Transformer 1% c64	412.50	476.34	1017.01	74/50000
Transformer 1% c128	851.98	970.86	2241.20	1612/50000
Shift external c1	10.27	11.58	12.42	0/43505

Warmup excluded; terminal latencies include failures. All individual throughput/counter records supplied.